

Embodied Presence Internet

Upload CSV

Rule-Based Insight · experimental prototype

Session P-002 showed high workload is paired with elevated error rates.

Refresh

MISSION CONTROL / ● LIVE

P-001

Remote object assembly

VR headset + haptic glove

68.8

rule-based stability indicator



4:14:04 PM / 119ms / heuristic stability 69% / rule flag: embodiment degradation

PDF-DERIVED EVALUATION FRAMEWORK

CATEM Cross-Layer Assessment

evidence-condition profile

Experience

69.3

Medium · 100% Coverage

Action

74

Medium · 100% Coverage

Human state & cognition

55.3

Medium · 100% Coverage

System conditions

68.6

Medium · 100% Coverage



Data & interpretation conditions

Cross-cutting record metadata · not an observational layer

81.5

high · 100% coverage

Metric coverage

100 %

Synchronization

83.2 %

Missing data

3.2 %

SELECTED EVIDENCE WINDOW

Robot object drop

-2s to +2s · synthetic fixture

Layer / measure	-2s	-1s	0s	+1s	+
EXPERIENCE					
Agency	78	72	58	64	7
ACTION					
Task error	0	0	1	0	
HUMAN STATE					
Workload	58 %	64 %	82 %	72 %	62
HUMAN STATE					
Heart rate	88 bpm	96 bpm	112 bpm	104 bpm	93
SYSTEM					
Latency	96 ms	118 ms	220 ms	154 ms	108
SYSTEM					
Packet loss	1.2 %	2.4 %	12 %	5.1 %	1.9

Layer / measure	-2s	-1s	0s	+1s	+
SYSTEM					
Tracking dropout	2 %	3 %	15 %	8 %	3

The five samples are deterministic authored values for software verification. Temporal co-occurrence does not establish causality.

CATEM Propositions P1–P6

live cross-layer tests

P1

MONITOR

Latency asymmetry

119 ms latency; agency 73; ownership 52

P2

MONITOR

Presence–performance dissociation

presence 74; workload 65; errors 8.9%

P3

MONITOR

Assistance trade-off

assistance 44; agency 73; trust 71

P4

MONITOR

Physiological leading indicators

HRV 50 ms; GSR 0.34; jitter 9 ms

P5

OBSERVED

Synchrony over fidelity

behavioral synchrony 72; avatar fidelity 66

P6

SUPPORTED

Ethical adaptation

disclosed True; overridable True; trust 71

Agency-Preserving Adaptation

explainable and overridable

Reduce rendering complexity and protect agency.

Continue monitoring workload.

Adaptation disclosure is active.

Experimental Dashboard Modules

future-work prototype

The modules below use deterministic rules or heuristic composites. They are not validated CATEM paper contributions, predictive models, diagnostic tools, or evidence of causal relationships.

Live Heuristic Streams

1 frames buffered

Latency

119ms



Rule-based stability indicator

69%



Heuristic embodiment estimate

45%



Rule-based risk signal

61%



Synthetic Telemetry Simulator

baseline alignment

Physiology

HR

94.43 bpm

HRV

50.09 ms

Stress

43.29 %

Network

Latency

118.94 ms

Jitter

9.2 ms

Loss

3.25 %

Embodiment

Agency

72.74 %

Ownership

Stress Escalation

Workload

64.97 %

Errors

51.63 %

Degrade

33.85 %

8.93 %

Timing

25.8 ms

Multimodal Presence Panel

partial-
presence

Presence

68.3 %

Intention

73.9 %

Emotion

78.6 %

Embodiment

44.9 %

Adaptive Support Panel

experimental · rule-
based · calm

Presence is holding. I will maintain sensory fidelity and monitor for drift.

retain current rendering settings

review network conditions

retain current haptic settings

Longitudinal Session Profile

76.1% continuity

P-001 adapts as a needs reinforcement with haptic reinforcement as the current recovery path.

Stability mode

performance

Preferred recovery

haptic reinforcement

Full-Sensory Telepresence

65% fidelity

Haptics

48.7 %

Visual

81.4 %

Audio

Tactile

79.8 %

50.1 %

Shared Reality Space

co-present

Trust

77.3 %

Team load

48.8 %

Social sync

77.3 %

Augment

75.1 %

Adaptation Control Panel

experimental · rule-based · stabilize

preserve embodiment continuity

retain current rendering settings

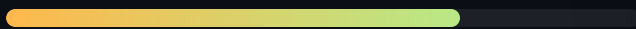
review network conditions

retain current haptic settings

rebalance sensory fidelity

Embodiment

71.7



Medium

Presence

75.5



High

Performance

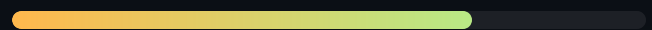
78.1



High

Behavior

72.4



Medium

Physiological

53.4



Medium

System

65.5



Medium

Visualization

50.5

Medium

Real-Time Multimodal Stream

real_telemetry_simulator

Latency

118.94 ms

Packet loss

3.25 %

Heart rate

94.43 bpm

Agency

72.74

Heuristic embodiment estimate

44.9

degraded

Cognitive State Heuristic

experimental · low
rule-based flag

Stability

68.8 %

Rule activation
index

36.2

Attention drift

29.2 %

Stress

nominal

Rule-Based Risk Review

experimental ·
descriptive threshold
logic

Nominal Input Condition

No event timing or probability is
estimated.

retain current rendering settings

review network conditions

retain current haptic settings

Deterministic threshold review only; no event time,
probability, predictive accuracy, diagnosis, or causal
effect is estimated.

Embodied-State Summary

experimental heuristic
· fragmenting

Heuristic Participant Profile

experimental heuristic ·
needs-reinforcement-
label

Awareness

71.1 %

Attention

70.8 %

Control

58.8 %

Continuity

70.2 %

Prototype fatigue indicator

44.2%

Latency sensitivity

high

Recovery strategy

haptic reinforcement

Reality Synchronization Engine

experimental composite · drifting

Robot

59.4 %

VR world

77.9 %

Body state

66.1 %

Unified

73.9 %

Narrative Summary Module

experimental · rule-
based · fragmenting

Heuristic state label: fragmenting. The nominal descriptive rule condition flags latency for review.

| presence continuity is 70.2%

| dominant explanatory signal: latency

| rule status: nominal

Risk Rule Monitor

1 active

Medium

embodiment degradation: agency or ownership is below the prototype rule threshold

Hypothesis Prompt Generator

rule-based prompt

Hypothesis: Visual-haptic mismatch may be associated with reduced ownership.

Test this using counterbalanced visual-haptic mismatch conditions with synchronized ownership ratings.

Session Pattern Summary

experimental heuristic · 76% rule-activation score

latency spike

ownership drift

adaptive compression

haptic reinforcement

Explanation Trace

experimental heuristic · 2 active factors

latency: latency exceeds the prototype rule threshold 32%

packet_loss: packet loss exceeds the prototype rule threshold 8%

Core Metrics

2026-05-22T10:00:00

Task time	130 s
Error rate	11 %
Heart rate	106 bpm
HRV	42 ms
Latency	82 ms
FPS	61
Packet loss	1.3 %

Relationships

2 sessions

Monitor

Candidate pattern: higher latency co-occurs with lower agency in current fixtures

Descriptive Pattern

Candidate descriptive pattern: higher embodiment and task performance co-vary in current fixtures

Descriptive Pattern

Higher workload ↔ more errors

Research Analytics

descriptive correlations with sample size

latency vs agency

insufficient n (2)

embodiment vs task_efficiency

insufficient n (2)

workload vs error_rate

insufficient n (2)

heart_rate vs error_rate

insufficient n (2)

Experimental Research Architecture

29 modules

- 1 human operator
- 2 VR/AR/robot interface
- 3 multimodal sensor fusion layer
- 4 real-time cognitive state engine
- 5 embodiment intelligence model
- 6 adaptive telepresence system
- 7 research analytics and AI copilot

Session and Fixture Runs

descriptive comparison

Session ID	Task	Latency	Errors	Prototype summary
P-001	Remote object assembly	118	6	69.3
P-002	Remote navigation	82	11	65.7